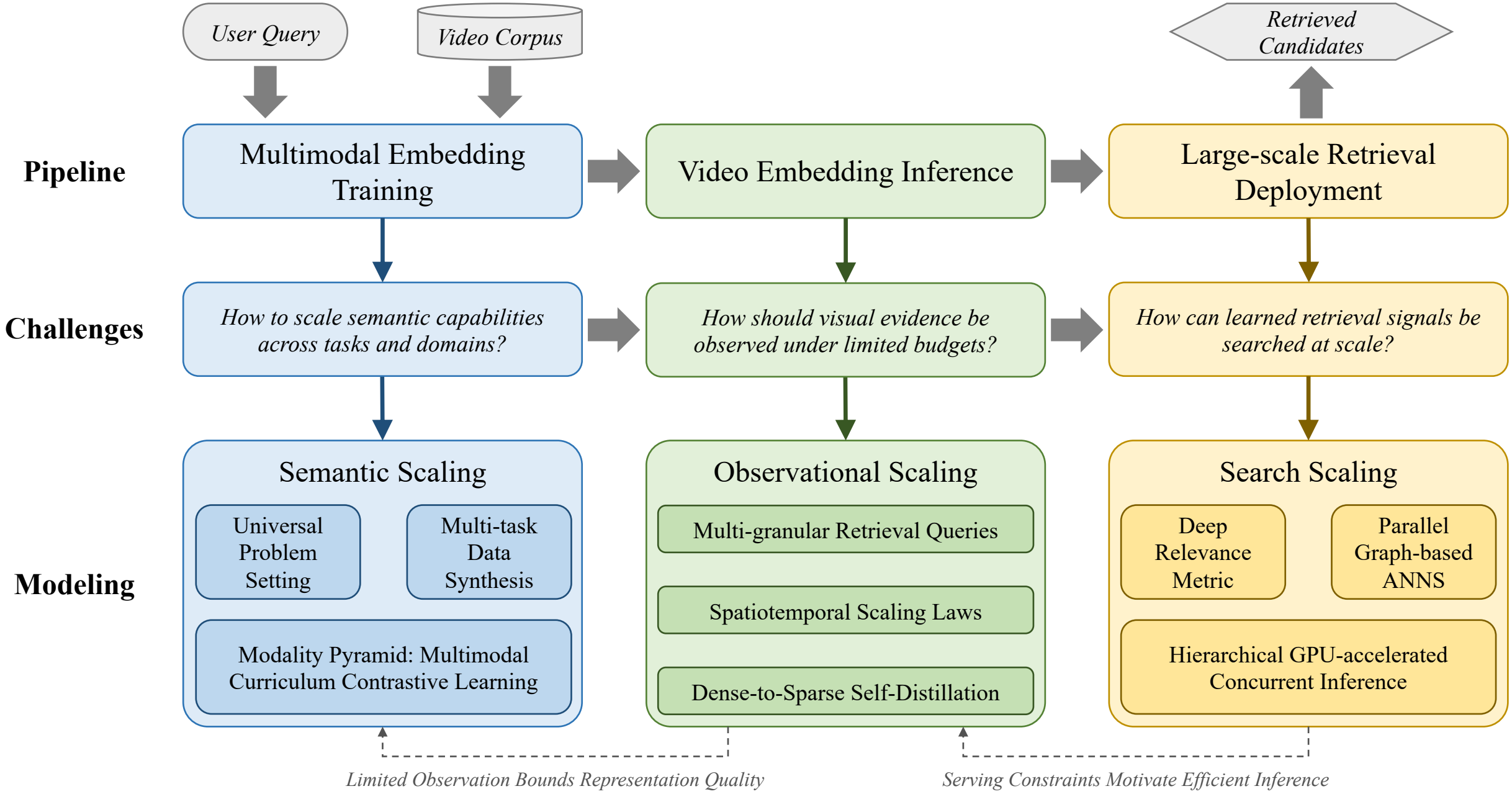

User Query
Video Corpus
Retrieved Candidates
Pipeline
Multimodal Embedding Training
Video Embedding Inference
Large-scale Retrieval Deployment
Challenges
How to scale semantic capabilities across tasks and domains?
How should visual evidence be observed under limited budgets?
How can learned retrieval signals be searched at scale?
Modeling
Semantic Scaling
Universal Problem Setting
Multi-task Data Synthesis
Modality Pyramid: Multimodal Curriculum Contrastive Learning
Observational Scaling
Multi-granular Retrieval Queries
Spatiotemporal Scaling Laws
Dense-to-Sparse Self-Distillation
Search Scaling
Deep Relevance Metric
Parallel Graph-based ANNS
Hierarchical GPU-accelerated Concurrent Inference
Limited Observation Bounds Representation Quality
Serving Constraints Motivate Efficient Inference